Blackmon Lab

Evolutionary Biology

Body Evolution in Arboreal vs. Non-arboreal Cricetidae: A Phylogenetic Comparative Analysis

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Background

The family Cricetidae represents one of the most speciose mammalian families, comprising over 600 species distributed across six continents, excluding Australia and Antarctica. This remarkable family includes hamsters, voles, lemmings, deer mice, pack rats, and New World rats and mice. Cricetids exhibit extraordinary ecological diversity, occupying habitats ranging from arctic tundra to tropical rainforests, from arid deserts to semi-aquatic environments. This diversity makes Cricetidae an ideal system for studying how ecological specialization shapes morphological evolution.

Within Cricetidae, ecological diversity includes both strictly terrestrial, fossorial, semi-aquatic, and arboreal species. Several genera have independently evolved arboreal habits. This variation provides an ideal natural experiment to test whether arboreality imposes constraints on body size evolution. Most cricetids are relatively small, typically ranging from 10 to 200 grams, raising the question of whether they may already exist below biomechanical thresholds where arboreal constraints become significant. This small size range contrasts sharply with larger mammals where arboreal constraints are well-established, making cricetids particularly interesting for understanding whether biomechanical principles scale consistently across body size ranges.

The study asks, do arboreal and non-arboreal cricetid lineages differ in patterns of body size evolution? If arboreality imposes biomechanical constraints, arboreal species should exhibit significantly smaller body sizes compared to their non-arboreal relatives, even after controlling for shared evolutionary history. Alternatively, if cricetids are already below critical size thresholds, no significant difference in body size should be observed between arboreal and non-arboreal lineages.

Study Design & AI

A time-calibrated phylogenetic tree containing 574 Cricetidae species was obtained from TimeTree [1]. Body mass measurements (grams) were extracted from the PanTHERIA database [2][3]. Each species was classified as arboreal or non-arboreal through systematic literature review using IUCN Red List, GBIF, and primary literature sources [2][3]. Species were classified as arboreal if they exhibited regular climbing behavior, nested in trees, or foraged in elevated vegetation. After matching species names between the tree and trait datasets, 306 species were retained for analysis: 40 arboreal and 266 non-arboreal.

Phylogenetic ANOVA was performed using the phylANOVA function in the phytools R package. This method tests for differences in body mass between groups while accounting for phylogenetic non-independence, where closely related species share traits due to common ancestry rather than independent evolution. The test used 1,000 permutation-based simulations to generate null distributions and provide p-values corrected for phylogenetic structure.

All analyses were conducted in R. Claude AI was used throughout this project to assist with R scripting, data cleaning, species name standardization, and figure production. All AI-generated code was reviewed and verified by the student and mentor. Biological classifications and data sources were independently researched and accessed [1][2][3].

Model Fitting

Phylogenetic ANOVA was performed using the phylANOVA function in the phytools R package to compare body mass between arboreal and non-arboreal cricetids. This method extends standard ANOVA by incorporating the phylogenetic tree structure into the analysis. Since closely related species share traits due to common ancestry rather than independent evolution, treating them as independent inflates sample size and produces false positives. The phylogenetic tree from TimeTree [1] accounts for this shared evolutionary history, isolating the true effect of lifestyle on body size.

The test uses 1,000 permutation-based simulations to generate null distributions under the hypothesis of no difference between groups, providing p-values corrected for phylogenetic structure. This approach ensures that detected differences reflect genuine evolutionary patterns rather than phylogenetic artifacts. The model tested was: Body Mass ~ Arboreal Status, with body mass as the continuous response variable and arboreal classification as the binary predictor.

Results

Phylogenetic ANOVA revealed no significant difference in log body mass between arboreal and non-arboreal cricetids when accounting for shared evolutionary history. The F-statistic was 0.096 with 1 and 402 degrees of freedom, yielding a p-value of 0.887. This result indicates that after controlling for phylogenetic relationships, habitat type does not predict body mass in Cricetidae. The observed difference in body mass between arboreal and non-arboreal species was not statistically distinguishable from patterns expected under a null model of no association between arboreality and body size.

Arboreal species were not phylogenetically clustered but were distributed across multiple subfamilies. Within Tylomyinae, arboreal genera include Tylomys, Ototylomys, and Nyctomys, which are specialized climbing rats. Within Neotominae, some Peromyscus species such as P. boylii and P. californicus exhibit arboreal behavior. Within Sigmodontinae, arboreal taxa include Rhipidomys, Oecomys, Juliomys, and Wiedomys. This phylogenetic dispersion suggests arboreality evolved multiple times independently within Cricetidae, strengthening the test's power to detect adaptive patterns if present.

Figure 1: Body mass distributions for arboreal and non-arboreal cricetids. Violin plots show density, with box plots indicating median and interquartile range. Sample sizes are n = 266 for non-arboreal and n = 40 for arboreal species. Distributions overlap strongly, and phylogenetic ANOVA shows no significant difference between groups.

Discussion

Contrary to initial predictions, this study found no evidence that arboreality constrains body size evolution in Cricetidae. After controlling for phylogenetic relatedness, arboreal and non-arboreal species did not differ significantly in body mass, with a p-value of 0.887. This result suggests that the biomechanical challenges of climbing do not impose strong directional selection for reduced body size in this rodent family. The absence of a significant relationship persists despite arboreal species being distributed across multiple independent lineages, which should have strengthened the analysis's power to detect adaptive convergence if it existed.

Most cricetids fall within a relatively narrow body size range of 10 to 200 grams, substantially smaller than taxa where arboreal size constraints are well-documented, such as primates ranging from 500 grams to 100 kilograms. Scaling relationships indicate that gravitational stress increases non-linearly with body mass, and cricetids may simply be too small for climbing biomechanics to impose significant selective pressure. Surface area-to-volume ratios at these small sizes may favor adhesion and grip strength regardless of habitat. This suggests that biomechanical thresholds for arboreal constraints may not apply uniformly across mammalian body size ranges.

Arboreal cricetids have evolved specialized morphological adaptations that may circumvent size constraints. Prehensile or semi-prehensile tails, present in genera such as Tylomys and Rhipidomys, provide additional support and balance during climbing. Enlarged foot pads and opposable digits enhance grip on branches. Relatively longer limbs and tails shift center of mass and improve stability during arboreal locomotion. Clawed digits provide mechanical advantage for clinging to vertical surfaces. These adaptations may decouple body size from climbing performance, permitting a broader range of body sizes in arboreal niches than predicted by simple biomechanical models that assume morphology remains constant.

With 40 arboreal species versus 266 non-arboreal species, the analysis had adequate power to detect moderate-to-large effect sizes but may have been underpowered for subtle differences. The observed 10.4 gram difference may represent a real but weak signal below the detection threshold. Alternatively, it may simply reflect sampling variation rather than a biological trend. The binary arboreal versus non-arboreal classification simplifies complex behavioral variation. A continuous arboreality index incorporating time spent in trees, nest height, foraging substrate use, and morphological specialization might reveal more nuanced patterns.

Conclusions & Acknowledgements

This study found no evidence for arboreal body size constraints in Cricetidae. Phylogenetic comparative analysis of 306 cricetid species revealed no significant difference in body mass between arboreal and non-arboreal lineages, with a p-value of 0.887. Cricetids may be below biomechanical thresholds where climbing imposes significant constraints. Morphological and behavioral flexibility may decouple body size from arboreal performance. Specialized adaptations including prehensile tails and enlarged foot pads, combined with semi-arboreal habits, may allow cricetids to climb effectively across a broad range of body sizes. Future studies should employ continuous arboreality metrics, morphometric analyses, and biomechanical measurements to test specific functional hypotheses about the relationship between body size and climbing performance.

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